# Road accidents in Italy: where to invest?

A road safety company wants to know where to invest.

I looked for the Italian towns with the most accidents: in total, per inhabitant and per km².

**168,557**

accidents per year

**3,075**

deaths per year

**28.6**

accidents per 10,000 inhabitants

Italy, average 2022-2024. Data: ISTAT (accidents) and SITUAS (population and area).

# What I did

1. Downloaded the ISTAT data with the API and saved it on my PC
2. Got population and area from SITUAS with web scraping
3. Cleaned the data and joined the two tables
4. Calculated accidents per 10,000 inhabitants and per km²
5. Grouped the towns with clustering (KMeans)
6. ANOVA test and a simple forecast (linear regression)
7. Built a dashboard in Power BI

## Problems I solved

- API limit: 5 calls per minute, so I saved the data once

- 2024: towns with 0 accidents were missing, I counted them as 0

- Very small towns have extreme rates: I used towns with 10,000+ inhabitants

# Tools

| Tool | Used for |
|---|---|
| **Python + pandas** | download, cleaning, new columns |
| **requests** | ISTAT API and SITUAS scraping |
| **matplotlib, seaborn** | charts |
| **scikit-learn** | clustering (KMeans) |
| **scipy, statsmodels** | ANOVA and regression |
| **Power BI** | dashboard |
| **Git, GitHub** | save and share the project |

# What I found




The drop in 2020 is because of the COVID lockdown.

- Most accidents per inhabitant: tourist towns and big cities like Bergamo, Florence, Genoa

- Most accidents per km²: Milan, Florence, Turin, Naples

- Centre and North have more accidents than South and Islands (ANOVA)

- 3 groups of towns: high risk per inhabitant, dense cities, fewer but more serious accidents

- Forecast: a small decrease, about 167,000 accidents in 2025

# Where to invest

### 1. High risk per inhabitant

326 towns, mostly Centre and North

Prevention and road safety checks

### 2. Dense cities

87 towns (Milan, Naples, Turin...)

Better traffic management

### 3. Fewer but more serious accidents

789 towns

Speed control on extra-urban roads

## Limits

- Only accidents with injured people or deaths
- Rates use residents only, so tourist towns look more dangerous

Project: github.com/aledoll01-sys/progetto_finale